1. An equation of the tangent line to the curve $y = \dfrac{x}{1-\ln(x-1)}$ at $x = 2$ is

$$y' = \frac{[1-\ln(x-1)] - x\left[\frac{-1}{x-1}\right]}{[1-\ln(x-1)]^2}$$

(a) $y = 3x - 6$

(b) $y = 3x + 7$

$$\text{slope} = y'(2) = \frac{1-\ln 1 + \frac{2}{1}}{(1-\ln 1)^2} = 3$$

(c) $y = 3x - 4$

$$\text{point } (2, y(2)) = (2, 2)$$

(d) $y = \frac{1}{3}x + 2$

$$\text{Eqn: } y - 2 = 3(x-2) \Rightarrow y = 3x - 4$$

(e) $y = -3x + 4$

2. The y-intercept of the tangent line to the graph of $y = \sin^{-1}\left(\frac{x}{2}\right)$ at $x = 1$ is

$$y' = \frac{\frac{1}{2}}{\sqrt{1-\frac{x^2}{4}}}, \quad \text{slope} = y'(1) = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}}$$

(a) $\frac{5\pi}{6} - \frac{\sqrt{3}}{3}$

$$\Rightarrow y' = \frac{\sqrt{3}}{3}. \quad \text{point } \left(1, \frac{\pi}{6}\right) \Rightarrow$$

(b) $\frac{2\pi}{3} - \sqrt{3}$

(c) $\frac{\pi}{3} - \sqrt{3}$

$$\text{eqn: } y - \frac{\pi}{6} = \frac{\sqrt{3}}{3}(x-1)$$

(d) $\frac{\pi}{6} - \frac{\sqrt{3}}{3}$

$$x = 0 \Rightarrow y = \frac{\pi}{6} - \frac{\sqrt{3}}{3}$$

(e) $\frac{\pi}{6} - \frac{1}{3}$

$$= y\text{-intercept}.$$

3. If $h(x)=\dfrac{1+xf(x)}{g(x)}$, $g(2)=1$, $g'(2)=3$, $f'(2)=5$, and $h'(2)=6$, then $f(2)$ is equal to

(a) -1

(b) $\frac{1}{5}$

(c) 5

(d) -5

(e) $\frac{2}{5}$

$$h'(x)=\frac{g(x)[f(x)+xf'(x)]-[1+xf(x)]g'(x)}{[g(x)]^2}$$

$$\Rightarrow h'(2)=\frac{g(2)[f(2)+2f'(2)]-[1+2f(2)]g'(2)}{[g(2)]^2}$$

$$\Rightarrow 6=\frac{(1)[f(2)+10]-[1+2f(2)](3)}{1}\Rightarrow$$

$$6=f(2)+10-3-6f(2)\Rightarrow -1=-5f(2)\Rightarrow$$

$$f(2)=\frac{1}{5}$$

4. If $f(x)=(1+x^{-1})^{-1}$, then $f'(x)=$

(a) 1

(b) $\frac{1}{(x+1)^2}$

(c) $-\left(\frac{x}{x-1}\right)^2$

(d) $\frac{x-1}{x}$

(e) $\frac{-1}{(x+1)^2}$

$$f'(x)=-(1+x^{-1})^{-2}(-x^{-2})$$

$$=\frac{-1}{\left(1+\frac{1}{x}\right)^2}\left(-\frac{1}{x^2}\right)$$

$$=\frac{1}{\left(\left(1+\frac{1}{x}\right)\cdot x\right)^2}=\frac{1}{(1+x)^2}$$

5. If $y \sin x = x^3 + \tan y$, then $\frac{dy}{dx} =$

(a) $\frac{y\cos x - 3x^2}{\sec^2 y - \sin x}$

(b) $\frac{y\sin x - 3x^2}{\sec^2 y - \sin x}$

(c) $\frac{3x^2}{2y\sec y^2 - \sin x}$

(d) $\frac{3x^2 - y\cos x}{\sec y - \sin x}$

(e) $\frac{3x^2}{\cos x}$

$\frac{d}{dx}(y\sin x) = \frac{d}{dx}(x^3 + \tan y) \Rightarrow$

$y\cos x + y' \sin x = 3x^2 + y' \sec^2 y.$

$\Rightarrow y\cos x - 3x^2 = y'(\sec^2 y - \sin x)$

$\Rightarrow y' = \frac{y\cos x - 3x^2}{\sec^2 y - \sin x}$

6. If $y = \sqrt[3]{x^4} - \frac{1}{\sqrt[4]{x^3}}$, then $\left.\frac{dy}{dx}\right|_{x=1} =$

(a) 0

(b) $\frac{7}{12}$

(c) 1

(d) $-\frac{3}{4}$

(e) $\frac{25}{12}$

$y = x^{4/3} - x^{-3/4}$

$y' = \frac{4}{3}x^{1/3} + \frac{3}{4}x^{-7/4}$

$y'(1) = \frac{4}{3} + \frac{3}{4} = \frac{25}{12}$

7. Let $f(x) = \cot(2x)$. Using the definition of the derivative, we get $f'\left(\frac{\pi}{4}\right) =$

(a) $\lim_{x\to\frac{\pi}{4}} \frac{4\cot(2x)}{4x-\pi}$

(b) $\lim_{h\to 0} \frac{\cot\left(\frac{\pi}{2}+h\right)}{h-\frac{\pi}{4}}$

(c) $\lim_{x\to\frac{\pi}{4}} \frac{\cot(2x)+1}{4x-\pi}$

(d) $\lim_{h\to 0} \frac{\cot\left(\frac{\pi}{4}+2h\right)-2}{h}$

(e) $\lim_{x\to\frac{\pi}{4}} \frac{\cot x - 1}{x-\frac{\pi}{4}}$

$$f'\left(\tfrac{\pi}{4}\right) = \lim_{x\to\pi/4} \frac{f(x)-f\left(\frac{\pi}{4}\right)}{x-\pi/4}$$

$$= \lim_{x\to\pi/4} \frac{\cot 2x - \cot\frac{\pi}{2}}{x-\pi/4}$$

$$= \lim_{x\to\pi/4} \frac{4\cot 2x}{4x-\pi}$$

8. If $y = xe^{cx}$, where c is a constant, then $y^{(4)} =$

(a) $c^3xe^{cx} + 2c^2e^{cx}$

(b) x^4e^{4cx}

(c) $c^4xe^{cx} + 4c^3e^{cx}$

(d) $4c^3xe^{cx} + c^3e^{cx}$

(e) $c^4xe^{cx} + 3c^3e^{cx}$

$$y' = e^{cx} + cxe^{cx}$$

$$y'' = ce^{cx} + ce^{cx} + c^2xe^{cx}$$

$$= 2ce^{cx} + c^2xe^{cx}$$

$$y''' = 2c^2e^{cx} + c^2e^{cx} + c^3xe^{cx}$$

$$= 3c^2e^{cx} + c^3xe^{cx}$$

$$y^{(4)} = 3c^3e^{cx} + c^3e^{cx} + c^4xe^{cx}$$

$$= 4c^3e^{cx} + c^4xe^{cx}$$

9. The value of the limit $\lim_{x\to 0} \frac{\sin x \tan(3x)}{x^3+2x^2}$

(a) is equal to $\frac{3}{2}$

(b) does not exist

(c) is equal to 3

(d) is equal to 0

(e) is equal to 1

$$= \lim_{x\to 0} \frac{\sin x \tan 3x}{x^2(x+2)}$$

$$= \lim_{x\to 0} \left(\frac{\sin x}{x}\right)\left(\frac{\tan 3x}{x}\right)\left(\frac{1}{x+2}\right)$$

$$= (1)(3)\left(\frac{1}{2}\right) = \frac{3}{2}.$$

10. If $f(x) = \frac{|x|}{\sqrt{2-x^2}}$, then $f'(-1)$

(a) does not exist

(b) is equal to -1

(c) is equal to 2

(d) is equal to -2

(e) is equal to 1

For $x<0 \Rightarrow$

$$f(x) = \frac{-x}{\sqrt{2-x^2}} \Rightarrow$$

$$f'(x) = \frac{\sqrt{2-x^2}(-1) - (-x)\frac{-2x}{2\sqrt{2-x^2}}}{(2-x^2)}$$

$$f'(-1) = \frac{(1)(-1) - (1)\frac{2}{2(1)}}{1}$$

$$= -1 - 1 = -2$$

11. If $f(x) = \log_{10} \sqrt[3]{\dfrac{x^2}{(x-1)^4}}$, then $f'(2) =$

(a) $\dfrac{-\ln 10}{3}$

(b) $\dfrac{-1}{3\ln 10}$

(c) $-2\ln 10$

(d) $\dfrac{-1}{\ln 10}$

(e) $\dfrac{-3}{\ln 10}$

$$f(x) = \frac{1}{3}\left[\log_{10} x^2 - \log_{10}(x-1)^4\right]$$

$$= \frac{2}{3}\log_{10} x - \frac{4}{3}\log_{10}(x-1)$$

$$f'(x) = \left[\frac{2}{3x} - \frac{4}{3(x-1)}\right]\frac{1}{\ln 10}$$

$$f'(2) = \left(\frac{1}{3} - \frac{4}{3}\right)\cdot\frac{1}{\ln 10}$$

$$= \frac{-1}{\ln 10}.$$

12. If $f(x) = \sqrt{x + \sqrt{3x + \sqrt{x}}}$, then $f'(1) =$

(a) $\dfrac{1}{\sqrt{3}}$

(b) $\dfrac{7}{8\sqrt{3}}$

(c) 3

(d) $\dfrac{5}{2\sqrt{3}}$

(e) $\dfrac{15}{16\sqrt{3}}$

$$f(x) = \left[x + \left(3x + x^{1/2}\right)^{1/2}\right]^{1/2} \Longrightarrow$$

$$f'(x) = \frac{1}{2}\left[x + (3x + x^{1/2})^{1/2}\right]^{-1/2}\left(1 + \frac{1}{2}(3x + x^{1/2})^{-1/2}\left(3 + \frac{1}{2}x^{-1/2}\right)\right)$$

$$f'(1) = \frac{1}{2}\left[1 + 4^{1/2}\right]^{-1/2}\left[1 + \frac{1}{2}(4)^{-1/2}\left(3 + \frac{1}{2}\right)\right]$$

$$= \frac{1}{2}\left[1+2\right]^{-1/2}\left[1 + \frac{1}{4}\cdot\frac{7}{2}\right]$$

$$= \frac{1}{2}\cdot\frac{1}{\sqrt{3}}\cdot\frac{15}{8} = \frac{15}{16\sqrt{3}}$$

13. The equation of motion of a particle moving in a straight line is given by

$$s(t) = t^3 - 12t + 3, \quad t \geq 0.$$

The particle is **speeding up** when

$v = 3t^2 - 12 = 3(t^2-4) = 3(t-2)(t+2)$

(a) $0 < t < 6$

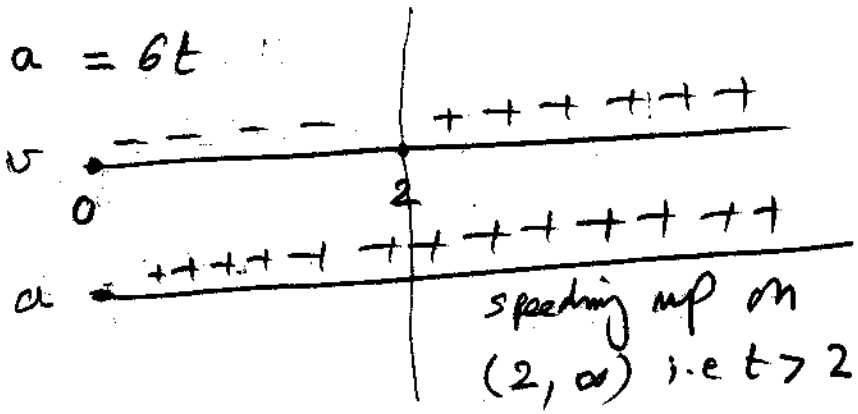
(b) $t > 1$

(c) $1 < t < 2$

(d) $0 < t < 2$

(e) $t > 2$

$a = 6t$



14. The function $f(x) = x^3(x^2 - 2x)^5$ has horizontal tangent lines at

(a) Six points

(b) Three points

(c) Four points

(d) Five points

(e) Two points

$f'(x) = 3x^2(x^2-2x)^5 + x^3\left[5(x^2-2x)^4(2x-2)\right]$

$= x^2(x^2-2x)^4\left[3(x^2-2x) + 5x(2x-2)\right]$

$= x^2(x^2-2x)^4(18x^2-16x)$

$= x^2 \cdot x^4 \cdot x(x-2)^2(18x-16)$

$= x^4(x-2)^2(18x-16)$

$f'(x) = 0$ at $x = 0, 2, \frac{16}{18}$

at 3 points.

15. The slope of the tangent line to the curve $2(x^2+y^2)^2 = 25(x^2-y^2)$ at the point $(3,1)$ is

(a) $\frac{9}{13}$

(b) $\frac{-9}{13}$

(c) -3

(d) $\frac{6}{13}$

(e) 0

$\frac{d}{dx} 2(x^2+y^2)^2 = \frac{d}{dx} 25(x^2-y^2) \Rightarrow$

$4(x^2+y^2)(2x+2yy') = 25(2x-2yy')$

at $(3,1)$, i.e, when $x=3$ and $y=1 \Rightarrow$

$4(10)(6+2y') = 25(6-2y') \Rightarrow$

$240 + 80y' = 150 - 50y' \Rightarrow$

$130y' = -90 \Rightarrow y' = -\frac{9}{13}$.

16. If the line $3x+y=b$ is tangent to the parabola $y=ax^2$ when $x=-2$, then $4a+b=$

(a) 1

(b) -2

(c) 0

(d) 3

(e) 5

$3x+y=b \Rightarrow y = -3x+b \Rightarrow$ slope $=-3$

$y = ax^2 \Rightarrow y' = 2ax \Rightarrow y'\big|_{x=-2} = -3$

$\Rightarrow -4a = -3 \Rightarrow \boxed{a = \frac{3}{4}}$

also at $x=-2 \Rightarrow -3x+b = ax^2$

i.e $6+b = 4a \Rightarrow 6+b = 4(\frac{3}{4}) = 3$

$\Rightarrow \boxed{b = -3}$

$\Rightarrow 4a+b = 4(\frac{3}{4}) - 3 = 3-3 = 0$

17. If $y = x^{99}(x+8)$, then $\frac{d^{100}y}{dx^{100}} =$

$y = x^{100} + 8x^{99}$

(a) 99!

(b) $99! + 8 \cdot (98!)$ If $y = x^n \Rightarrow y^{(n)} = n!$

(c) 0

If $y = x^{n-1} \Rightarrow y^{(n)} = 0$

(d) 100

(e) 100! $\Rightarrow y = x^{100} + 8x^{99} \Rightarrow$

$y^{(100)} = 100! + 0 = 100!$

18. If $\frac{d}{dx}[f(2x)] = x^2$, then $f'(6) =$

(a) $\frac{9}{2}$ $\frac{d}{dx}[f(2x)] = x^2 \Rightarrow$

(b) $\frac{1}{2}$ $f'(2x) \cdot 2 = x^2 \Rightarrow$

(c) 6 $f'(2x) = \frac{x^2}{2}$

(d) 3

(e) 9 $x = 3 \Rightarrow f'(6) = \frac{9}{2}$

19. If $y = x - 2\tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$, then $\frac{dy}{dx} =$

(a) $\frac{1}{(1+\cos x)^2}$

(b) $\frac{2}{1+\cos x}$

(c) 0

(d) $\frac{\cos x}{1+\cos x}$

(e) 1

$$y' = 1 - 2\,\frac{\frac{(1+\cos x)(\cos x) - (\sin x)(-\sin x)}{(1+\cos x)^2}}{1 + \frac{\sin^2 x}{(1+\cos x)^2}}$$

$$= 1 - 2\,\frac{\cos x + \cos^2 x + \sin^2 x}{(1+\cos x)^2 + \sin^2 x}$$

$$= 1 - 2\,\frac{\cos x + 1}{1 + 2\cos x + \cos^2 x + \sin^2 x}$$

$$= 1 - \frac{\cos x + 1}{\cos x + 1} = 1 - 1 = 0$$

20. If $y = \left[\sin\left(\frac{1}{x}\right)\right]^x$, then $\frac{xy'}{y} =$

(a) $-\csc\left(\frac{1}{x}\right) + x\ln\left(\sin\left(\frac{1}{x}\right)\right)$

(b) $\frac{1}{x}\cot\left(\frac{1}{x}\right) + \ln\left(\sin\left(\frac{1}{x}\right)\right)$

(c) $x^2\csc\left(\frac{1}{x}\right) + x\ln\left(\sin\left(\frac{1}{x}\right)\right)$

(d) $\ln\left(\sin\left(\frac{1}{x}\right)\right) - \cot\left(\frac{1}{x}\right)$

(e) $x\ln\left(\sin\left(\frac{1}{x}\right)\right) - \cot\left(\frac{1}{x}\right)$

$$\ln y = x \ln \sin\left(\tfrac{1}{x}\right)$$

$$\frac{y'}{y} = \ln\left(\sin\tfrac{1}{x}\right) + x\,\frac{\left(-\frac{1}{x^2}\right)\cos\left(\frac{1}{x}\right)}{\sin\left(\frac{1}{x}\right)}$$

$$\frac{y'}{y} = \ln\left(\sin\tfrac{1}{x}\right) - \frac{1}{x}\cot\left(\tfrac{1}{x}\right)$$

$$\Rightarrow \frac{xy'}{y} = x\ln\left(\sin\tfrac{1}{x}\right) - \cot\left(\tfrac{1}{x}\right)$$